

Lecture 5 2026-09-15

Last time: linear least squares

Today: N.C.O. for equality constrained optimization, least norm

Case 3: $y = Ax$ when A is wide

$$\boxed{A} \quad A \in \mathbb{R}^{m \times n} \\ m < n$$

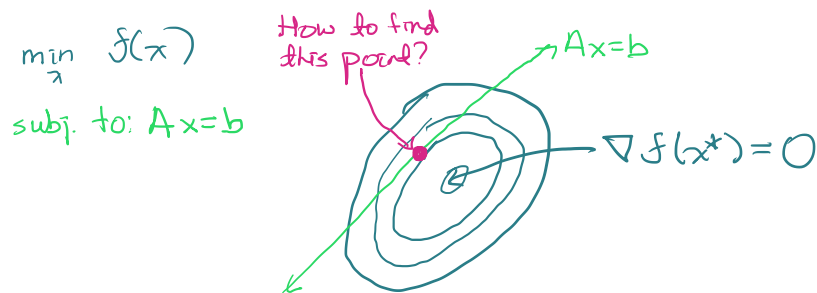
$y = Ax$ corresponds to going from
high-dimensional $x \in \mathbb{R}^n$ to low
dimensional $y \in \mathbb{R}^m$

What are necessary conditions of optimality

$$\begin{array}{l} \text{for} \quad \min_x f(x) \\ \text{subj. to } Ax = b \end{array}$$

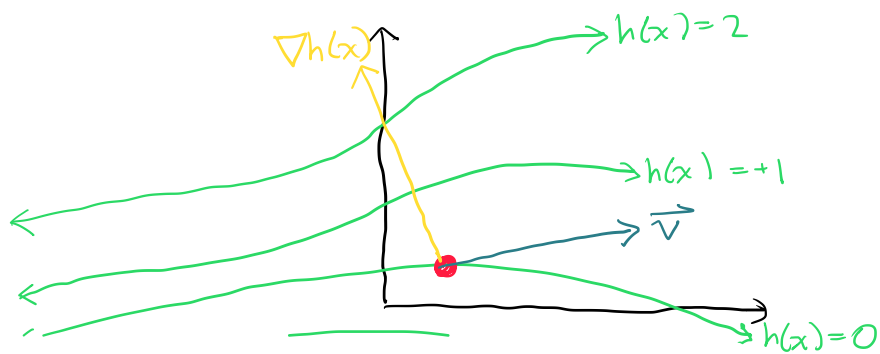
Recall n unconstrained case:

if x^* is local minimizer, then $\nabla f(x^*) = 0$



Generalize to: $\min_x f(x)$
 subj. to: $h(x)=0$

How can we move along $-\nabla f(x^{(k)})$ while satisfying $h(x^{(k)})=0$?





$\vec{v}$: points in tangent direction of $h(x)=0$

$\nabla h(x)$: points in normal direction of $h(x)=0$

1. Taking steps along local tangent $\vec{v}$ will keep $\nabla f(x^{(k)})$ on constraint surface $h(x^{(k)})=0$

2. Taking steps along direction $\pm \nabla h(x^{(k)})$ will violate constraint $h(x)=0$

We can represent $\nabla f(x^{(k)})$ as:

$$\nabla f(x^{(k)}) = \mu \vec{v} + \lambda \nabla h(x^{(k)})$$

where μ, λ are some scalars

When implementing gradient descent, if we start with some feasible solution $x^{(0)}$ such that $h(x^{(0)})=0$, then we can ensure constraint remains satisfied by only taking steps along $\mu \vec{v}$

Thus, our new stopping condition for implementing gradient descent in the equality constrained case is that we iterate until there is no valid step in the $\mu \vec{v}$ direction, i.e.,:

$$\nabla f(x^*) = \cancel{\mu} \vec{v} + \lambda \nabla h(x^*)$$

$\mu = 0$

$$\nabla f(x^*) = \lambda \nabla h(x^*) \quad \text{stopping}$$

$$\rightarrow \nabla f(x^*) = \lambda \nabla h(x^*) \Rightarrow \begin{array}{c} \text{stopping} \\ \text{condition} \end{array} : \nabla f(x^*) + \lambda \nabla h(x^*) = 0$$

$\uparrow$
 moved $\lambda \nabla h(x^*)$
 for convenience

What if we have multiple constraints $h_i(x) = 0$?

$\rightarrow$ Each constraint $\nabla h_i(x^*)$ has an associated λ_i

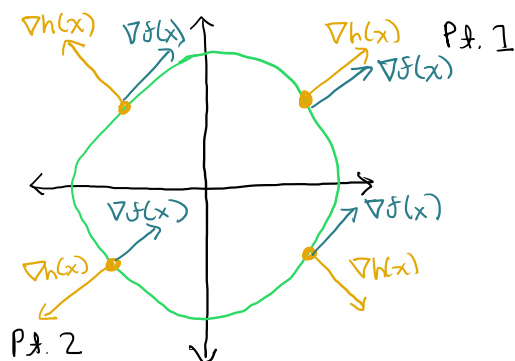
$$\Rightarrow \nabla f(x^*) + \sum_{i=0}^{n_{eq}} \lambda_i \nabla h_i(x^*) = \nabla f(x^*) + \underbrace{\lambda^T \nabla h(x^*)}_{\text{stacked vector of constraints}} = 0$$

Example: $\min_x x_1 + x_2$

subj. to: $x_1^2 + x_2^2 - 4 = 0$

$$f(x) = x_1 + x_2 \rightarrow \nabla f(x) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$h(x) = x_1^2 + x_2^2 - 4 \rightarrow \nabla h(x) = \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix}$$



Both points 1 and 2 are candidate local minimizers as there exists some λ such that $\nabla f(x) + \lambda \nabla h(x) = 0$ at both points

Note: these are only necessary conditions of optimality, not sufficient $\rightarrow$ we can clearly see Pt. 2 is the true minimizer

Putting this altogether:

$$\begin{array}{ll} \min_x f(x) & f \in \mathcal{C}^2 \\ \text{subj. to: } h(x) = 0 & h \in \mathcal{C}^2 \end{array}$$

Necessary conditions of optimality: if x^*

is a local minimizer, then

$$1. \quad h(x^*) = 0$$

$$2. \quad \nabla f(x^*) + \lambda \nabla h(x^*) = 0$$

Lagrangian: suppose we integrate expression

2 w.r.t. x :

$$\mathcal{L}(x, \lambda) = f(x) + \lambda^T h(x) \quad \left(+ \text{ some integration constant that we'll ignore} \right)$$

Then we can succinctly write N.C.O as:

$$1. \quad \nabla_{\lambda} \mathcal{L}(x^*, \lambda^*) = 0$$

$$2. \quad \nabla_x \mathcal{L}(x^*, \lambda^*) = 0$$

Least norm problem:

$$\min_x \frac{1}{2} x^T x$$

$$\text{subj. to: } Ax = b$$

$$\rightarrow \mathcal{L}(x, \lambda) = \frac{1}{2} x^T x + \lambda^T (Ax - b)$$

Let's apply N.C.O.: $\nabla_x \mathcal{L}(x^*, \lambda^*) = 0$

$$\nabla_{\lambda} \mathcal{L}(x^*, \lambda^*) = 0$$

$$\mathcal{L} = \frac{1}{2} x^T x + \lambda^T (Ax - b)$$

$$\Rightarrow \nabla_{\lambda} \mathcal{L} = Ax - b := 0$$

$$\nabla_x \mathcal{L} = x + A^T \lambda := 0 \Rightarrow x^* = -A^T \lambda^*$$

$$\Rightarrow Ax^* - b = 0 = A(-A^T \lambda^*) - b = 0$$

$$\Rightarrow \lambda^* = -(AA^T)^{-1} b$$

$$x^* = -A^T \lambda^* = A^T (AA^T)^{-1} b$$

$$\Rightarrow \text{solution to } \min_x \frac{1}{2} x^T x \quad \Rightarrow x^* = A^T (AA^T)^{-1} b$$

s.t. $Ax = b$